

Dataset-specific retrieval

prompt_1 = "How to make Chocolate Frango Mints?"

print("LLM: ", llm_chain.invoke(prompt_1))

print("RAG: ", rag_chain.invoke(prompt_1))

print("Advanced RAG: ", advanced_rag(prompt_1))

LLM: Ingredients:

- 1 cup chocolate chips
- 1/4 cup heavy cream
- 1/2 teaspoon peppermint extract
- 1/2 cup powdered sugar
- Green food coloring (optional)
- 1/4 teaspoon vanilla extract

Instructions:

1. In a small saucepan, melt the chocolate chips and heavy cream over low heat, stirring constantly until smooth.
2. Remove the saucepan from the heat and stir in the peppermint extract and vanilla extract. If desired, add a few drops of green food coloring for a minty green color.
3. Gradually stir in the powdered sugar until well combined and smooth.
4. Pour the mixture into a shallow dish lined with parchment paper and refrigerate for about 1 hour, or until firm.
5. Once the mixture is firm, use a small cookie cutter or knife to cut out small squares or rectangles.
6. Store the chocolate frango mints in an airtight container in the refrigerator until ready to serve.

Enjoy your delicious homemade chocolate frango mints!

RAG: To make Chocolate Frango Mints, mix confectioners sugar, butter, water, and mint until stiff.

Then, form the mixture into patties before dipping them in chocolate and placing them on wax paper to set. Enjoy your homemade Chocolate Frango Mints!

Advanced RAG: To prepare Homemade Chocolate Frango Mints, mix all ingredients until stiff, form into patties, then dip in chocolate and place on wax paper to set. The ingredients you will need include confectioners sugar, butter, water, and mint.

Inexistent recipe identification

prompt_2 = "Do you have a recipe for Sichuan peppercorn and soy sauce infused chocolate mousse?"

print("LLM: ", llm_chain.invoke(prompt_2))

print("RAG: ", rag_chain.invoke(prompt_2))

print("Advanced RAG: ", advanced_rag(prompt_2))

LLM: Here is a recipe for Sichuan peppercorn and soy sauce infused chocolate mousse:

Ingredients:

- 200g dark chocolate (70% cocoa)
- 1 cup heavy cream
- 2 tbsp Sichuan peppercorns
- 2 tbsp soy sauce
- 3 egg yolks
- 2 tbsp sugar
- Pinch of salt
- Sea salt flakes, for garnish

Instructions:

1. In a small saucepan, heat the heavy cream over medium heat until it just begins to simmer. Add the Sichuan peppercorns and soy sauce, then remove from heat and let steep for 10 minutes.
2. Strain the cream mixture to remove the peppercorns, then return the mixture to the saucepan and heat again until simmering.
3. In a separate bowl, whisk together the egg yolks, sugar, and a pinch of salt. Slowly pour the hot cream mixture into the egg yolk mixture, whisking constantly.
4. Return the mixture to the saucepan and cook over low heat, stirring constantly, until it thickens slightly and coats the back of a spoon.
5. Remove from heat and add the chopped dark chocolate, stirring until melted and smooth.
6. Pour the mixture into individual serving dishes and refrigerate for at least 2 hours, until set.
7. Before serving, sprinkle with sea salt flakes for a delicious finishing touch.

Enjoy your Sichuan peppercorn and soy sauce infused chocolate mousse!

RAG: No, I do not have a recipe for Sichuan peppercorn and soy sauce infused chocolate mousse.

Advanced RAG: I don't know.

Creativity and synthesis

prompt_3 = "Based on the recipes in the dataset, suggest a full three-course meal that maintains a cohesive flavor profile. Provide reasonings for the choices."

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print("LLM: ", llm_chain.invoke(prompt_3))
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print("RAG: ", rag_chain.invoke(prompt_3))
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print("Advanced RAG: ", advanced_rag(prompt_3))
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LLM: Starter: Caprese Salad

- The Caprese Salad with fresh mozzarella, ripe tomatoes, and fragrant basil is a classic Italian dish that sets a light and refreshing tone for the meal. The combination of creamy cheese, juicy tomatoes, and aromatic herbs provides a well-balanced and flavorful start to the meal.

Main Course: Chicken Alfredo Pasta

- Chicken Alfredo Pasta is a rich and indulgent dish that complements the lightness of the Caprese Salad. The creamy Alfredo sauce coats the tender chicken and pasta, creating a decadent and satisfying main course. The flavors of the Parmesan cheese and garlic in the Alfredo sauce tie in well with the basil in the Caprese Salad, creating a cohesive flavor profile for the meal.

Dessert: Tiramisu

- Tiramisu is a classic Italian dessert that rounds out the meal with its delicate layers of coffee-soaked ladyfingers and creamy mascarpone mixture. The hint of coffee flavor in the dessert complements the savory notes of the Chicken Alfredo Pasta, while the light and airy texture of the Tiramisu provides a refreshing end to the meal. The overall flavor profile of the three-course meal is cohesive, with each dish complementing and enhancing the others.

RAG: For a cohesive meal, you can start with Anchoiade as an appetizer, followed by Pasta with Crispy Prosciutto and Arugula Pesto as the main course, and finish with Memes Meal for dessert. The Anchoiade and Pasta dishes both have herbaceous flavors that complement each other, while the Memes Meal provides a sweet and savory end to the meal.

Advanced RAG: A cohesive three-course meal could include roasted butternut squash soup as an appetizer, lemon garlic herb roasted chicken with vegetables as the main course, and vanilla bean crème brûlée as the dessert. The flavors of the roasted butternut squash soup will provide a creamy and earthy start, followed by the zesty and herbal notes of the roasted chicken complemented by the richness of the vanilla bean crème brûlée for a satisfying finish. Each dish offers a unique flavor profile that combines well to create a balanced and enjoyable meal.

Cuisine classification and reasoning

prompt_4 = "Provide me with a spicy Asian cuisine for dinner."

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print("LLM: ", llm_chain.invoke(prompt_4))
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print("RAG: ", rag_chain.invoke(prompt_4))
```

```
print("Advanced RAG: ", advanced_rag(prompt_4))
```

LLM: Spicy Szechuan Beef with stir-fried vegetables and steamed rice.

RAG: Cajun Sea Scallops would be a spicy Asian cuisine option for dinner. The dish involves sauteing scallops with Cajun seasoning, hot sauce, and garlic in a skillet. It offers a flavorful and spicy seafood option for a meal.

Advanced RAG: An option similar to a Spicy Szechuan Stir-Fry dish would be a Hot And Spicy Asian Marinade, which includes ingredients like soy sauce, sesame oil, garlic, and ginger. Another choice could be Spicy Asian Dumpling Sauce with soy sauce, lemon juice, garlic, and green chili peppers for a kick of spice. The Asian Dipping Sauce with chili paste, sesame oil, and wasabi is also a flavorful option for a spicy Asian dinner.

Dataset-specific variations

prompt_5 = "How does the chocolate cake recipe in your dataset differ from a traditional one?"

State the names of the recipes identified."

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print("LLM: ", llm_chain.invoke(prompt_5))
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print("RAG: ", rag_chain.invoke(prompt_5))
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print("Advanced RAG: ", advanced_rag(prompt_5))
```

LLM: The chocolate cake recipe in my dataset is called "Decadent Chocolate Cake" and it differs from a traditional chocolate cake recipe in a few ways:

1. The recipe includes coffee as one of the ingredients, which can enhance the chocolate flavor of the cake.
2. The recipe also calls for buttermilk, which can add moisture and a slight tanginess to the cake.
3. The frosting for the cake is a rich chocolate ganache, which is a bit more decadent than a traditional buttercream frosting.

Let me know if you would like me to provide the full recipe for the "Decadent Chocolate Cake" in my dataset.

RAG: The Spago'S Chocolate Chiffon Cake recipe differs from a traditional one by using oil and separating egg whites for folding into the batter. The Torta Alla Gianduia (Chocolate Hazelnut Cake) recipe deviates by incorporating Nutella and ground hazelnuts into the batter. The Craig Claiborne'S Rum Chocolate Dessert recipe stands out with the addition of rum and heavy cream for a rich texture.

Advanced RAG: The unique ingredients in the "Avocado Chocolate Cake" recipe are avocados, and in the "Coconut Sugar Chocolate Cake" recipe are coconut sugar, which differentiate them from traditional chocolate cake recipes in the dataset.